

20 Polymerisation

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Addition polymerisation joins alkene monomers without small-molecule by-products. Condensation polymerisation links bifunctional monomers and removes small molecules, commonly water or HCl. Hydrolysable ester and amide links make some condensation polymers biodegradable; properties depend on structure, crystallinity and cross-linking.

Worked example: follow the reasoning

$n \text{ CH}_2=\text{CHCl} \rightarrow [-\text{CH}_2-\text{CHCl}-]_n$; $\text{C}=\text{C}$ becomes a saturated chain while side group is retained.

Quick check · multiple choice

A What small molecule is often lost during polyester formation? [1]

- A O_2
- B N_2
- C Br_2
- D H_2O

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Identify polymerisation type for ethane-1,2-diol and ethanedioic acid, and name the linkage and small molecule released. [3]

Answers and explanation

Quick check: D. H_2O

Question 1 [3 marks]

Condensation [1]; ester link [1]; water [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

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